Grade 7
Unit 13 Assessment

Name____________________
Date_________________
Period__________

**Showing Work with a Calculator:** Because you are using a calculator, it is possible to accidently punch the wrong button. To ensure you earn partial credit, in each work box show the operation(s) that you entered into the calculator and the result.

1. A real estate agent tracked the prices of homes sold in a specific neighborhood. The stem-and-leaf plot of the home prices, in thousands, for the homes sold in the neighborhood is shown.

| **Key** | $18|3 = \$183,000$ |
|---|---|

**Home Price**

| **Stem** | **Leaf** |
|---|---|
| 9 | 1 |
| 10 | |
| 11 | |
| 12 | |
| 13 | |
| 14 | |
| 15 | |
| 16 | |
| 17 | 1, 2, 5, 8 |
| 18 | 0, 3, 7 |
| 19 | 2, 4 |

Complete the statement. Show your work.

a. The measure of center that should be used to represent the typical sale price of the homes in the neighborhood is the

- ○ mean
- ○ median

home price. The typical price for the neighborhood is $____________.

work

b. The data

- ○ has an outlier,
- ○ is generally symmetric,

so

- ○ the range is
- ○ the interquartile range is
- ○ both the range and the interquartile range are

the preferred measure(s) of variation.

2. The annual lightning report gives the lightning density, the number of lightning strikes per square kilometer (strikes per $km^2$), for each US state. A researcher uses the information in the report to compare the lightning strikes in the southern US states and in the western US states. The researcher's data is shown.

| **Southern US States** | | **Western US States** | |
|---|---|---|---|
| **State** | **Lightning Density (strikes/$km^2$)** | **State** | **Lightning Density (strikes/$km^2$)** |
| **Alabama** | 37.87 | **Arizona** | 12.10 |
| **Arkansas** | 41.09 | **California** | 1.03 |
| **Florida** | 85.99 | **Colorado** | 12.80 |
| **Georgia** | 35.95 | **Montana** | 5.19 |
| **Louisiana** | 81.62 | **Nevada** | 2.55 |
| **Mississippi** | 55.79 | **New Mexico** | 16.24 |
| **Missouri** | 42.60 | **Oregon** | 0.77 |
| **Oklahoma** | 54.91 | **Utah** | 5.72 |
| **Tennessee** | 35.87 | **Washington** | 0.30 |
| **Texas** | 60.25 | **Wyoming** | 6.17 |

Complete each statement. Show your work. If necessary, round to the nearest hundredth.

a. The southern states have a median of ___________ strikes/$km^2$ and a

mean of ___________ strikes/$km^2$.

| work |
|---|
| |

b. The western states have a median of __________ strikes/km$^2$ and a mean of __________ strikes/km$^2$.

| work |
|---|
| |

c. Neither data set has an outlier. Therefore, the

- ○ mean
- ○ median

should represent the center and the

- ○ range
- ○ interquartile range

should be used to represent variability.

d. The western states tend to have

- ○ less
- ○ more

variability than the southern states because the

- ○ range
- ○ interquartile range
- ○ range and the interquartile range

is/are

- ○ larger
- ○ smaller

than the

- ○ range
- ○ interquartile range
- ○ range and the interquartile range

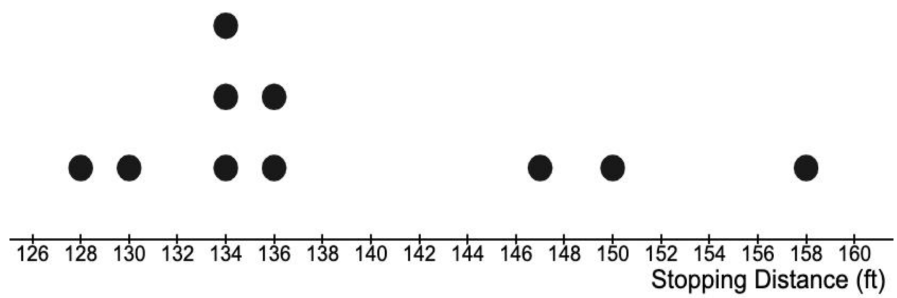
for the southern states.

3. A tire manufacturer is testing the braking performance of one of its tire models on a test track. The company evaluated the tires on $10$ different cars, recording the stopping distance, in feet (ft), for each car driving $50$ miles per hour on both wet and dry pavement. The data is shown.

**Stopping Distance on Wet Pavement**

| Car | Stopping Distance on Wet Pavement (ft) |
|---|---|
| ***A*** | 201 |
| ***B*** | 220 |
| ***C*** | 192 |
| ***D*** | 146 |
| ***E*** | 182 |
| ***F*** | 173 |
| ***G*** | 202 |
| ***H*** | 180 |
| ***K*** | 192 |
| ***L*** | 206 |

**Stopping Distance on Dry Pavement**




Complete the statements.

a. A car's stopping distance on dry pavement is typically
- ○ less
- ○ more

than the stopping distance on wet pavement. The stopping distance on dry pavement has
- ○ less
- ○ more

variability than the stopping distance on wet pavement.

b. The stopping distance of car ***D*** on wet pavement is
- ○ a gap
- ○ a cluster
- ○ an outlier

which means that the
- ○ mean
- ○ median

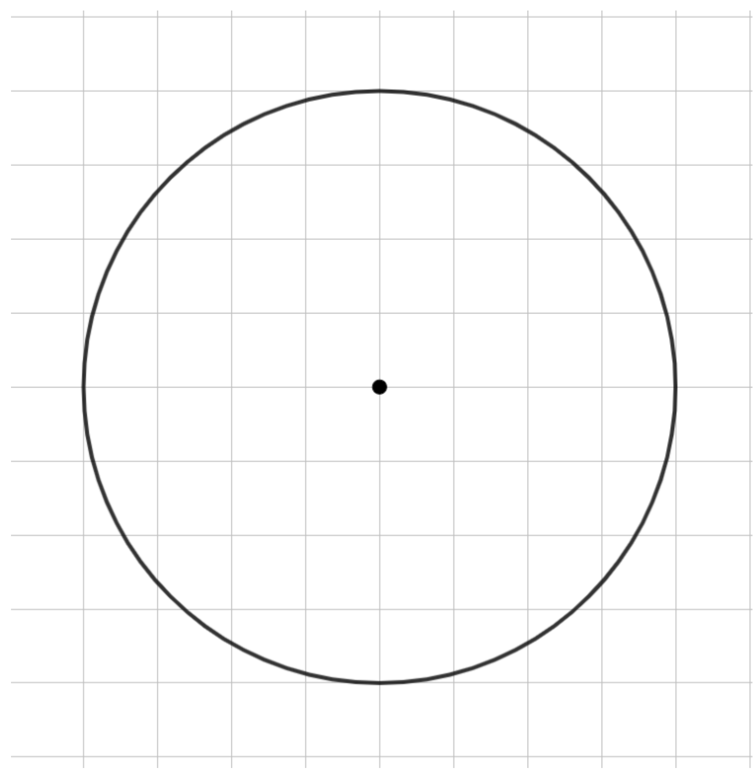
is the best measure of center and the
- ○ range
- ○ interquartile range

is the best measure of spread for this data.

4. A survey was conducted to determine food choices of the $80$ students at a school picnic. The picnic food choices are shown in the table.

| Picnic Food | Number of Students |
|---|---|
| **Hamburger** | 12 |
| **Hot Dog** | 20 |
| **Pizza** | 24 |
| **Salad** | 8 |
| **Sandwich** | 16 |

a. Create a circle graph to represent this data. Show your work

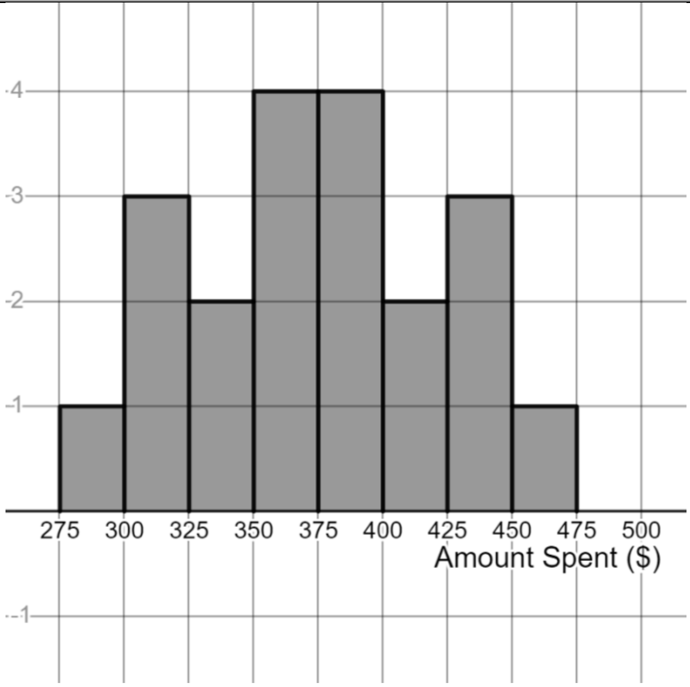
| Work | Picnic Food |
|---|---|
| |  |

b. Select all of other graphical representations that would be appropriate for the data.

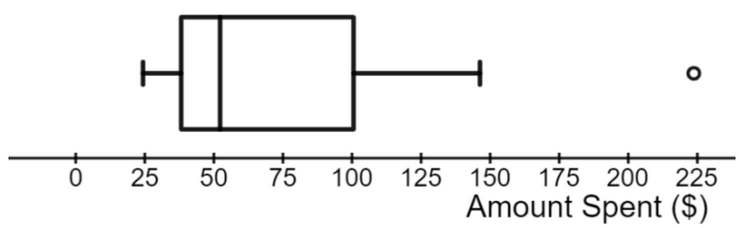
- ☐ Box plot
- ☐ Bar graph
- ☐ Frequency table
- ☐ Histogram
- ☐ Stem-and-Leaf plot

5. A financial analyst studied the shopping habits of 20 shoppers in December and January. The histogram displays the amount of money, in dollars ($), the shoppers spent in December and the boxplot displays the amount of money, in dollars ($), the same shoppers spent in January.

**December**




**January**




Complete the statement.

The data displays show that the distribution of the amount spent, in dollars, in December is

- ○ less than
- ○ more than
- ○ approximately the same as

the amount spent, in dollars, in January.